

TechXNinjas

Presents

paradox 2.0

A 3-Months National-Level Innovation Hackathon

BUILD · PITCH · WIN

SECURE-CERT

Team Name:

Track:

Problem Statement:

Source Code

Web3

Certificate Fraud is a \$4 Billion Annual Problem.

Problem Statement & Proposed Solution

Today's certificates are centralized, prone to forgery, and slow to verify, which creates significant trust issues.

Key Issues:

- **Easy Forgery:** Physical and digital certificates can be easily tampered with or fabricated.
- **Slow Verification:** Manual processes can take days or weeks, delaying hiring and admissions.
- **Lack of Transparency:** Centralized systems are like black boxes with no way to independently verify authenticity.

Impact: 40% of resumes contain false educational credentials, and fake licenses pose serious public safety risks.

Problem Statement & Proposed Solution

Our Solution: SECURE-CERT (A Decentralized System for Verifiable Digital Trust.)

SECURE-CERT is a blockchain-based system that makes certificate verification tamper-proof, transparent, and instant.

Core Features:

- **Immutable Storage:** Certificate hashes are stored permanently on the Ethereum blockchain.
- **Instant Verification:** Authenticity can be confirmed in under 3 seconds without special software.
- **Privacy-First:** Only cryptographic hashes are stored on the blockchain, not the certificate content itself, preserving privacy.
- **No Intermediaries:** Direct verification eliminates the need for costly third-party services.

Working Prototype & Flow of Solution

Fully Functional Web App Deployed on Sepolia Testnet.

Prototype Link Deployed on pythonanywhere.com:

➡ <https://sourcecode.pythonanywhere.com/> ⬅

A Simple Two-Step Process for Issuers and Verifiers.

Step 1: Issue

- An organization uploads a certificate file (PDF, JPG, PNG) and enters a unique ID.
- The system generates a unique SHA-256 hash and stores it on the blockchain via a MetaMask transaction.

Step 2: Verify

- Anyone enters the certificate ID and uploads the corresponding file.
- The system re-calculates the hash and compares it to the record on the blockchain, providing an instant "AUTHENTIC" or "TAMPERED/INVALID" result.

Working Prototype & Flow of Solution

Our prototype is a mobile-responsive Progressive Web App (PWA) that works on all devices.

Key Implemented Features:

- Certificate Issuance and Instant Verification.
- Real-time blockchain integration with transaction tracking on Etherscan.
- Intuitive user flow for both technical and non-technical users.

Screenshots:

Homepage: <https://drive.google.com/file/d/1aXjMWOIVPY-u5PDQMwe9IarvCLk8Kz8E/view?usp=sharing>

Issue Certificate: https://drive.google.com/file/d/1uzW_8yyPE6EYT4RKfgTLmbycFds-YEew/view?usp=drive_link

Verify Certificate: https://drive.google.com/file/d/1Ank50ZBK0ojXuTlae0bNB-3lmvclbcqF/view?usp=drive_link

Help & Support: https://drive.google.com/file/d/1nzq3fD8ZVl0zVguyHxsjQxWjrFWp2pC1/view?usp=drive_link

Contact Information: https://drive.google.com/file/d/1yjWDAXUohTXbKMTyK7bl0ikT1ysQomDr/view?usp=drive_link

Design & UI/UX

User-Centric Approach

- The interface is designed with simplicity in mind, ensuring that both technical and non-technical users can issue and verify certificates without confusion.
- **Visual Trust Indicators:** Blockchain connectivity status, real-time network information, and cryptographic proof displays

Accessibility & Usability

- **Mobile-First Design:** Responsive design adapts seamlessly across desktop and mobile devices. Clean layouts, intuitive navigation, and clear call-to-action buttons make interactions effortless.
- **Intuitive Workflow:** Clear step-by-step process with visual feedback at each stage

Innovation & Uniqueness

Unique Selling Proposition:

SECURE-CERT is the first fully decentralized verification system that requires no registration, accounts, or special software to verify a certificate.

Novel Architecture:

We use a hybrid approach of SHA-256 hashing to ensure privacy combined with blockchain for immutability.

Competitive Advantages:

- No Recurring Fees (only blockchain gas costs).
- No Vendor Lock-In and no central authority required.
- Permanent Records that exist as long as the Ethereum network does.

Feasibility & Impact

- **Target Audience:** Educational institutions, employers, professional licensing bodies, and individuals seeking portable credentials.
- **Technical Feasibility:** Built on mature, battle-tested technologies like Ethereum and Flask with a scalable architecture.
- **Economic Impact:** Has the potential to save organizations over \$100M annually in verification costs and reduce time-to-hire by 70%.
- **Social Impact:** Reduces educational fraud, enables trustless remote hiring, and democratizes credential verification globally.

Technologies Used (Tech Stack)

Built on a Modern, Secure, and Scalable Tech Stack.

- **Frontend:** Flask, Bootstrap 5, JavaScript ES6+, Web3.js.
- **Backend:** Python Flask, Web3.py, Gunicorn.
- **Blockchain:** Ethereum (Sepolia testnet), Solidity, MetaMask, Infura.
- **Security:** SHA-256 hashing, Input Validation, and CORS protection.

Clarity & Relevance

Social Need:

- **Educational Institutions:** Need cost-effective way to issue tamper-proof diplomas and certificates
- **Employers:** Require fast, reliable method to verify candidate credentials
- **Professional Bodies:** Must ensure licensing integrity for public safety
- **Individuals:** Want portable, verifiable credentials that work globally

Solution Alignment:

Our blockchain-based approach directly addresses each pain point with a modern, scalable solution that eliminates intermediaries while providing cryptographic proof of authenticity.

Research, References & Future Scope

Research Foundation:

- "Blockchain Technology for Education: A Systematic Review" - IEEE Access 2021"
- Digital Credentials and Blockchain: State of the Art" - Journal of Network and Computer Applications
- Ethereum Yellow Paper - Technical specifications for blockchain implementation
- W3C Verifiable Credentials Data Model - Standards for digital credentials
- Lovnish Verma, Budhiraja, A., Singh, S. (2024). Blockchain-Based Certificate Verification System: A Decentralized Approach. In: NICEDT 2024. Lecture Notes in Networks and Systems, vol 1022. Springer, Singapore.

https://doi.org/10.1007/978-981-97-3601-0_36

Future Roadmap:

- Phase 1 (Next 3 Months): Deploy on Mainnet, develop RESTful APIs for enterprise integration, and launch a mobile app.
- Phase 2 (6 Months): Introduce multi-chain support (e.g., Polygon) and create a dashboard for bulk certificate operations.
- Phase 3 (12 Months): Integrate AI-powered fraud detection and decentralized identity(DID) solutions.
- Long-Term Vision: To become the global standard for digital credential verification, enabling a trustless economy